No. of Printed Pages : 4

Roll No. ....

220331/180331/170342
/120331/031731/032233
/030332/105232

3rd Sem / Automobile, Mechanical Engg., Mecatronics, Prod, T&D, Plastic, GE, CNC, CAD/CAM, Found. & Forg, Metallurgy, Print Making Tech., Mech (Ad. Manu. Tech.), Mech Engg (Fabrication Tech), Rubber Tech, Polymer Tech, AME, Mech. Engg. (Prod.)

Subject : Strength of Materials / Mechanics of Solids Basic Mech. Engg.

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The unit of stress is- (CO1)

a) Newton b) Pascal

c) N/cm d) KN/m

Q.2 Neutral axis always passes through (CO3)

a) Centroid of the beam through

b) Middle of the beam section

c) Centre of the beam

d) None of the above

Q.3 Effect of compressive force is to______ the length of body. (CO1)

a) Decrease b) Increase

c) Same d) None of above

Q.4 The centre of area of a plane lamina will not lie its geometrical centre is a (CO4)

a) Rectangle

b) Square

c) Right angled-triangle

d) Circle

Q.5 Initial radius of curvature of a given lead spring is kept depending upon (CO8)

a) Maximum load on the string

b) Maximum deflection of the string

c) Its proof load

d) Kind of its use

Q.6 The assumption made in Euler's column theory is that (CO7)

a) The failure of column occurs due to buckling alone

b) The length of column is very large as compared to its cross-sectional dimensions

c) The column material obeys Hooke's law

d) All of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 A beam whose_______ Ends are fixed is called fixed beam. (CO3)

Q.8 Define working stress. (CO1)

Q.9 Define slenderness ratio.

Q.10 Define Hook's law? (CO1)

Q.11 Define spring Index.

Q.12 Write down the units of strain.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain stress-strain diagram for ductile materials. (CO1)

Q.14 Explain assumption made in theory of simple bending. (CO5)

Q.15 What is the difference between strut & Column? (CO6)

Q.16 Define any five properties of material. (CO1)

Q.17 Define M.O.I. Write the formula for M.O.I of a circle about horizontal & vertical axis passing through the centroid? (CO3)

Q.18 Explain Euler's theory for long columns. (CO6)

Q.19 Define springs. Also explain different types of springs in brief. (CO5)

Q.20 Explain the various end conditions in the column. (CO6)

Q.21 Give the classification of load.

Q.22 Define Helical Springs and Name the two important type of springs. (CO8)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 A simply supported beam 6m long is subjected to 3 point loads of 2KN, 2KN and 4.5 KN each at a distance of 2m, 3m and 4m respectively from the left hand support. Draw the SFD & BMD for the beam. (CO5)

Q.24 Drive the expression for pure bending equation for a beam. (CO5)

Q.25 Find the M.O.I of T-section (20x3) cm & (20x3) cm about its horizontal & vertical centroidal axes. (CO4)